Xiaoyu Wang

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github.com/wxystudio | scholar.google.com/citations?user=Emze7GAAAAAJ

EDUCATION

Ph.D. in Biological Systems Engineering *01/2023 – 05/2027 (Expected)*
University of Wisconsin–Madison, Madison, WI
Minor: Computer Sciences
Relevant Coursework: CS 533 (Image Processing); CS 760 (Machine Learning); CS 759 (High Performance Computing for Applications in Engineering)

Doctoral Studies in Computing and Information Sciences *08/2022 – 01/2023*
Rochester Institute of Technology, Rochester, NY
Golisano College of Computing and Information Sciences
Relevant Coursework: CSCI 820 (Quantitative Foundations); CSCI 865 (Deep Learning)

B.S. in Computer Science and Technology *09/2017 – 06/2021*
Xi'an Jiaotong University, Xi'an, China

EXPERIENCE

Graduate Research Assistant *01/2023 – Present*
University of Wisconsin–Madison, Madison, WI

- Built and deployed an interactive geospatial platform for exploring county-level corn and soybean yield predictions across multiple years and ten in-season and end-of-season forecasting dates, with maps of predicted yield, error, and model uncertainty.
- Implemented county search, multi-county historical comparisons between model predictions and USDA NASS yields, uncertainty visualization, filtered CSV exports, and retrieval of NASS yield, planted-area, and harvested-area data.
- Operationalized on-demand yield inference for selected counties and user-drawn map regions with queued job tracking; integrated a Vue 3/MapLibre GL JS frontend with FastAPI model and data services, and released the live Web Tool and source code.

Research Intern *04/2021 – 04/2022*
Microsoft Research Asia, Media Computing Group, Beijing, China
Supervised by: Xiangyu Kong and Xiulian Peng

- Designed a multi-modal multi-correlation learning framework for audio-visual speech separation that explicitly modeled speaker-identity and phonetic correlations using contrastive and adversarial training.
- Integrated a U-Net separation backbone with wav2vec 2.0, lip-reading, and ResNet-18 speaker-identification models in PyTorch; trained and evaluated models on four NVIDIA V100 GPUs.
- Improved LRS3 SDR from 8.46 to 9.98, PESQ from 2.27 to 2.584, and STOI from 0.843 to 0.861 without increasing inference complexity; published a first-author paper at Interspeech 2022.

Research Intern *09/2020 – 03/2021*
SenseTime, Autonomous Driving Group, Shanghai, China
Supervised by: Guohang Yan and Tao Ma

- Contributed to OpenCalib, an open-source multi-sensor calibration toolbox for autonomous driving, by implementing calibration workflows among LiDAR, radar, and camera sensors.
- Trained a deep learning model to determine whether multi-sensor calibration results were accurate, automating calibration-quality assessment.

Remote Research Intern *06/2020 – 09/2020*
Nanyang Technological University
Supervised by: Tao Bai and Jun Zhao

- Proposed a data-free targeted universal adversarial perturbation method with a custom loss that balanced model disruption and target-label objectives without access to training data.
- Extended the attack to federated-learning settings and published the work as a first-author paper at SciSec 2021.

Research Intern *06/2019 – 02/2020*
INNNO

- Built a remote-sensing change-detection pipeline for drone imagery, including image alignment, bi-temporal image preprocessing, and semantic-segmentation mask prediction.
- Created synthetic change pairs to address limited labeled data and trained a Python-based model to detect changes in buildings, waterways, and vegetation; released the project code.

Research Assistant *01/2019 – 09/2019*
College of Artificial Intelligence, Xi'an Jiaotong University
Supervised by: Jinjun Wang

- Developed Distributed-GAN, enabling multiple users to train generative models locally on private data without sharing raw datasets while improving generated-sample diversity.
- Published the work as a first-author arXiv preprint.

Research Assistant *10/2017 – 2018*
Xi'an Jiaotong University
Supervised by: Jinsong Han
Conducted research on computer security and deep learning.

PUBLICATIONS

Journal Articles

Assimilation of LAI and FAPAR within DSSAT Guided by Leaf Growth Dynamics to Enhance Maize Yield Phenotyping 2026
Zezhong Tian et al.
Computers and Electronics in Agriculture, vol. 250, article 111946, 2026.

Learning County from Pixels: Corn Yield Prediction with Attention-Weighted Multiple Instance Learning 2025
***Xiaoyu Wang** et al.*
International Journal of Remote Sensing, vol. 46, no. 7, pp. 2815–2845, 2025.

Book Chapter

Satellite-Based Mapping of Crop Yield 2026
Yuchi Ma et al.
In Agricultural Applications of Earth Observation, ch. 7, pp. 193–230, Elsevier, 2026.

Conference Papers

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions 2025
Zhengwei Yang et al.
Accepted to IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025.

County-Level Crop Yield Prediction Using SMAP-Derived Data Products and Deep Learning Model 2024
Zhengwei Yang et al.
IEEE International Geoscience and Remote Sensing Symposium (IGARSS), pp. 3222–3225, 2024.

Multi-Modal Multi-Correlation Learning for Audio-Visual Speech Separation 2022
***Xiaoyu Wang**, Xiangyu Kong, Xiulian Peng, Yan Lu*
Interspeech, 2022.

A Data-Free Approach for Targeted Universal Adversarial Perturbation 2021
***Xiaoyu Wang**, Tao Bai, Jun Zhao*
Science of Cyber Security (SciSec), pp. 126–138, 2021.

Preprint

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions 2025
***Xiaoyu Wang** et al.*
arXiv:2503.16328, 2025.

TEACHING EXPERIENCE

Teaching Assistant, BSE 301: Land Information Management *Fall 2026*
University of Wisconsin–Madison
Revise lecture slides and answer student questions to support course learning.

AWARDS & HONORS

Stars of Tomorrow Excellent Intern Award, Microsoft Research Asia, 2021.
BSE Travel Award, University of Wisconsin–Madison, 2025. ($1,000)
Third-Class Award, Xi'an Jiaotong University, 2017. (Top 20% by GPA)

PROFESSIONAL SERVICE

Journal Reviewer
International Journal of Applied Earth Observation and Geoinformation (JAG)

TECHNICAL SKILLS

Languages:	Python, C++, Shell, MATLAB, CUDA C
Machine Learning & GPU:	PyTorch, TensorFlow, TensorRT, CUDA, cuDNN, cuBLAS
Web & Deployment:	FastAPI, Vue 3, MapLibre GL JS, Vite, Gunicorn, Docker
GIS & Remote Sensing:	Google Earth Engine, GDAL, QGIS, ArcGIS
Vision, Audio & Robotics:	OpenCV, librosa, Asteroid, Kaldi, FFmpeg, PCL, ROS
Development:	Linux, Git, CMake